

Proteus: Adaptive Scale-Space Analysis for Learning Non-Linear Fuzzy Manifold Memberships (*Draft*)

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Abstract

We present Proteus, a two-stage framework for learning explicit, interpretable, and generative models of high-dimensional data supported on non-linear manifolds across multiple nested scales. Proteus separates fast geometric discovery from statistically grounded refinement: Stage 1 performs a scale-space search with an augmented Growing Neural Gas to produce a multiscale scaffold that supports efficient querying and coarse classification. Stage 2 lifts this scaffold to a dynamic simplicial complex and refines it into a piecewise-linear generative core, optionally composing localized invertible warps where curvature exceeds the capacity of a linear approximation. Discrete structural edits (splits, merges/prunes, and warp attachments) are proposed by geometric diagnostics but accepted only when they improve a local free-energy proxy derived from finite-data transition statistics, providing an evidence-based notion of model selection under noise and weak sampling. Finally, a dual-flow reconstruction yields a divergence-consistent probability field on the learned complex, enabling likelihood evaluation, semantic queries under fuzzy constraints, and conditional synthesis within a single representation.

1 Introduction

Real-world data rarely lies near a single linear subspace; it is often concentrated on non-linear manifolds with spatially varying intrinsic dimension and structure that is only visible at particular scales. A useful model must therefore do more than produce a descriptive embedding: it should (i) *discover* characteristic scales without extensive tuning, (ii) *represent* the geometry explicitly enough to be interpretable, and (iii) *support inference* tasks such as likelihood evaluation and conditional generation. The current state of the art is not a single solution to this full stack, but a fragmented toolchain: embedding methods can reveal structure for visualization, TDA provides topological diagnostics, and probabilistic generative models can fit densities, yet these components typically do not yield a unified, explicit geometric representation with an evidence-based mechanism for discrete structure selection. Likewise, graph/mesh-growth approaches can construct explicit scaffolds efficiently in many regimes, but they are often governed by heuristics for stopping and complexity control, making it difficult to separate genuine unresolved structure from variance due to noise or weak sampling.

Proteus aims to reduce this gap by learning an explicit, multiscale geometric representation together with an evidence-based rule for discrete structure selection. It exposes a single representation that supports three modes of interaction: *conditional synthesis* from partial or fuzzy

constraints, *semantic querying* of data regions (e.g., density, anomaly, or manifold-membership queries under constraints), and *constraint extraction* that summarizes global structure from selected data subsets.

Architecturally, Proteus follows a coarse-to-fine, two-stage pipeline. Stage 1 performs fast scale-space-guided exploration using an augmented Growing Neural Gas to build a multiscale scaffold that captures coarse adjacency, topology, and mass partitioning at candidate scales. Stage 2 lifts this scaffold to a simplicial complex and refines it into a generative core, optionally augmenting locally with invertible warps where curvature requires additional capacity. Throughout, Proteus follows the principle of “geometry proposes, evidence disposes”: geometric diagnostics propose discrete structural edits, but an edit is accepted only when it improves a local free-energy proxy derived from finite-data transition statistics, providing an operational notion of model selection under noise.

Contributions.

- A two-stage framework that separates scalable multiscale discovery (Stage 1) from evidence-grounded geometric refinement into a generative model (Stage 2).
- A scale-space controller for characteristic-scale search that enables hierarchical decomposition where supported by data evidence.
- A simplicial generative representation with a local evidence test for discrete edits, unifying splits/merges/pruning and optional localized warps under a single acceptance criterion.
- A dual-flow reconstruction that yields a divergence-consistent probability field on the learned complex for likelihood queries and conditional synthesis.

Roadmap. Section 2 details the Proteus framework. Section 3 evaluates model quality and ablations, with derivations and implementation details provided in the Supplementary Information (SI).

1.1 Related Work

Proteus synthesizes concepts from manifold learning, computational geometry, and probabilistic modeling, occupying a unique niche at their intersection.

Manifold Learning and TDA. Classical manifold learning techniques, such as Isomap, LLE, t-SNE, and UMAP, excel at revealing low-dimensional structure within high-dimensional data. However, these methods primarily yield coordinate embeddings rather than explicit, generative geometric models; they are descriptive rather than constructive. Similarly, Topological Data Analysis (TDA), particularly persistent homology, provides powerful diagnostics for shape characterization but does not inherently produce a generative mechanism. Proteus bridges this gap by constructing an explicit simplicial complex that is both topologically faithful and generatively capable.

Generative Deep Learning. Modern generative models like VAEs, GANs, and Normalizing Flows have achieved remarkable success in density estimation and synthesis. Yet, they often operate as “black boxes,” obscuring the topological structure of the learned distribution. Proteus offers an interpretable alternative: a “white box” model where the data density is represented by a piecewise-linear manifold (with local corrections). While we utilize invertible flows for local warping, they are employed as targeted upgrades to the geometric scaffold rather than the primary modeling substrate.

Self-Organizing Maps and GNG. Our work builds directly upon the tradition of Self-Organizing Maps (SOMs) and Growing Neural Gas (GNG). While these algorithms are effective for vector quantization and topology learning, they historically lack a rigorous statistical objective

for model selection and termination. Proteus augments the GNG formalism with scale-space theory to guide growth and a variational free-energy objective to arbitrate structural complexity, transforming a heuristic clustering process into a principled statistical estimator.

Discrete Geometry and Physics. The theoretical underpinnings of our refinement stage draw from Discrete Exterior Calculus (DEC). We treat the simplicial complex not just as a mesh, but as a discrete domain for field theories. Our “Torsion Ladder” and “Dual Flow” mechanisms are discrete analogs of smooth operators, allowing us to enforce physical constraints—such as divergence consistency—directly on the learned manifold.

Predictive Coding and Free Energy. Finally, Proteus aligns with the Free Energy Principle (FEP) found in neuroscience and statistical physics. The “geometry proposes, evidence disposes” mechanic can be viewed as a form of structural predictive coding, where geometric stress (torsion, variance) generates prediction errors that are resolved either by continuous parameter updates (mass settling) or discrete topological interventions (splits/merges), minimizing a unified variational free energy.

2 The Proteus Framework

2.1 At a Glance

Proteus learns an explicit geometric representation of a high-dimensional data distribution together with an evidence-based rule for adding or removing structure. The key design choice is a two-stage separation between (i) fast multiscale exploration that discovers candidate structure cheaply and (ii) high-fidelity refinement that adds geometric capacity only where supported by finite-data evidence. Stage 1 performs a scale-space search and builds a hierarchy of coarse scaffolds; Stage 2 lifts a selected scaffold to a simplicial complex at a fixed characteristic scale and refines it into a generative core.

Learned artifacts. Given samples $X \subset \mathbb{R}^d$, Proteus produces:

- a multiscale *scaffold* (a hierarchy of node–edge graphs with associated local statistics) and characteristic scale choices τ^* from Stage 1;
- a refined simplicial complex (Stage 2) that serves as a piecewise-linear generative core at a fixed scale τ^* ;
- a reconstructed probability field on the complex for likelihood/density evaluation and sampling, optionally composed with localized invertible warps where curvature requires non-linear capacity.

Unifying learning rule: “geometry proposes, evidence disposes.” Proteus unifies its components with a simple control loop: (i) *propose* candidate discrete edits using fast geometric diagnostics (e.g., variance, torsion, and element quality), (ii) *test* each edit using a local free-energy proxy computed from finite-data transition statistics and a complexity penalty (SI S3.4, SI S13), and (iii) *settle* continuous parameters between accepted edits. Convergence at a fixed scale is defined operationally by equilibration of local statistics, after which the final topology is used to reconstruct a probability field for inference and generation.

2.2 Stage 1: Fast Exploration and Scaffolding

Stage 1 is the framework’s fast “front end”: it builds a coarse geometric scaffold quickly, and it does so in a way that produces actionable signals for where additional structure might be warranted. Conceptually, Stage 1 can be read as four coupled functional components.

(i) State: local statistical summaries. Stage 1 represents the data at a candidate scale by a set of nodes and their induced Voronoi regions, forming a *distributional scaffold* (a node–edge graph) that approximates a maximum-entropy partition. Rather than relying on heuristic error counters, each node i maintains exponential moving averages of the first two moments of its local residual distribution: the mean ($\mathbf{m}_i$) and squared error ($\mathbf{s}_i$). These summaries induce derived quantities used throughout Stage 1, including a local variance (σ_i^2) and a signal incoherence score (ρ_i). Each node also tracks a principal error direction ($\mathbf{u}_i$) via Oja’s rule, providing a data-driven axis along which to propose geometric expansion. The estimator memory is tied to neighborhood size by setting the EWMA weight to $\alpha = \ln 2/k$ (SI S2.2).

(ii) Per-sample update: routing evidence into the scaffold. Given a sample, Stage 1 identifies the k nearest nodes and distributes the learning signal with rank-ordered geometric weights $(1/2, 1/4, 1/8, \dots)$. This defines a discrete exponential neighborhood kernel and yields stable fractional “hit” statistics that can be reused for pruning and scale evaluation (details in SI S2.3).

(iii) Structure adaptation: propose edits, then gate them by evidence. The role of the statistics above is not only to move nodes but to propose targeted discrete edits. In Stage 1, a variance exceedance ($\sigma_i^2 > \tau_{local,i}$) proposes that a region may contain unresolved probability mass; a new node is then proposed along the principal error direction. Complementarily, weakly supported nodes/edges are proposed for pruning. Because high variance can also arise from noise or weak evidence, these discrete proposals are accepted only when they improve the local fit to observed transition statistics after a complexity penalty (SI S3.4), implementing the “geometry proposes, evidence disposes” rule at the scaffold level.

(iv) Stabilization and handoff: deciding when a scale is “done.” Stage 1 continues until local statistics equilibrate at the current scale, detected by a small-tolerance stabilization criterion (default $CV < 0.01$ on neighbor-normalized incoherence; SI S2.3 and SI S3). The resulting scaffold and its routed evidence are then passed to the meta-level scale-space controller below, which searches over candidate thresholds and recurses into subregions only when supported by sufficient routed data.

Computationally, per-sample updates are $\mathcal{O}(kd)$, with occasional deferred $\mathcal{O}(d^2)$ work on moves/splits; nearest-neighbor queries are handled with HNSW. Full pseudocode and implementation details are provided in SI S11.

2.2.1 Principled Scale Representation

We employ a four-tier representation of scale to decouple the optimizer’s search from the GNG’s local, adaptive physics (see SI S2.4 for the control-to-threshold mapping).

1. **Control Parameter** ($s_{control}$): A normalized parameter $\in [0, 1)$ proposed by the optimizer.
2. **Subspace Dimensionality** ($D_{subspace}$): The dimensionality of the current sub-problem.
3. **Global Growth Threshold** (τ_{global}): A physical scale derived from the control parameter via $\tau_{global} = -D_{subspace} \log(1 - s_{control})$, where $D_{subspace}$ is the dimensionality of the current problem. This global threshold is used to evaluate the overall quality of the learned structure at a given scale.
4. **Local Growth Threshold** ($\tau_{local,i}$): A per-node threshold, $\tau_{local,i} = -d_{final,i} \log(1 - s_{control})$, where $d_{final,i}$ is the smoothed, local intrinsic dimension at node i . This allows the GNG to adapt its growth conditions to regions of varying manifold dimensionality.

2.2.2 Scale-Space Grid, Optimizer, and Recursive Search

Classical scale-space theory studies how structures appear and disappear under Gaussian smoothing at scale σ , using scale-normalized operators to make responses comparable across scales. In Proteus, the Stage 1 growth threshold τ plays an analogous role: it bounds local residual variance, so increasing τ yields coarser partitions (with $\tau \equiv \sigma^2$ under a calibration; SI S2.1). We define a scale-space response $\Phi(\tau)$ from routed fractional hits and a local k -NN volume proxy, which behaves like a kernel density estimate under local stationarity (up to constants). With a one-time calibration $\sigma = \sqrt{\tau}/c_{d,k}$ and choosing $\gamma=1$ for scale-covariance, peaks of the normalized response in $\log \tau$ identify characteristic scales (SI S2.1).

We maximize $\Phi(\tau)$ over a geometric grid of thresholds $\{\tau_i\}$. The grid spacing follows a scale-space resolution limit: adjacent samples in $\log \sigma$ are separated by half the FWHM of a normalized Gaussian response, yielding a geometric ratio $r \approx 1/\sqrt{2} \approx 0.71$ (SI S2.1). At each τ_i , Stage 1 is run to its stabilization criterion (default $\text{CV} < 0.01$), and $\Phi(\tau_i)$ is recorded. Candidate maxima are bracketed via sign changes in a discrete second difference over $\log \tau$, then refined with a Gaussian-process Bayesian optimizer restricted to the bracket to obtain τ^* .

The procedure then recurses on the data partitions induced by the converged mesh at τ^* , discovering nested structure only where sufficient routed data exists (minimum sample thresholds as in SI S9). The meta-level controller is summarized below.

Algorithm 1 Recursive Scale-Space Search (meta-level controller)

```

1: procedure FINDSCALES( $X$ )
2:   Choose geometric grid  $\mathcal{T} = \{\tau_0 r^j\}_{j=0}^J$  with  $r=1/\sqrt{2}$ 
3:   for each  $\tau \in \mathcal{T}$  do
4:     Initialize mesh  $\mathcal{M} \leftarrow \emptyset$ ; repeat  $\mathcal{M} \leftarrow \text{UpdateWithSample}(\mathbf{x}, \mathcal{M}, \tau)$  until  $\text{CV} < 0.01$ 
5:     Record response  $\Phi(\tau)$ 
6:   end for
7:    $\mathcal{B} \leftarrow \text{BracketsFromSecondDifference}(\Phi, \mathcal{T})$ 
8:    $\tau^* \leftarrow \text{BayesianOptimize}(\Phi, \mathcal{B})$ 
9:    $\mathcal{M}^* \leftarrow \text{RunStage1ToEquilibrium}(X, \tau^*)$ ;  $\{X_k\} \leftarrow \text{PartitionByMesh}(X, \mathcal{M}^*)$ 
10:  for each partition  $X_k$  do
11:    if  $|X_k| \geq n_{\min}$  then
12:      FINDSCALES( $X_k$ ) ▷ Recurse on sufficiently large sub-regions
13:    end if
14:  end for
15: end procedure

```

The result of Stage 1 at a single scale is a scaffold: a node–edge graph that captures manifold adjacency and mass partitioning at the chosen threshold. The full hierarchical scaffold is obtained by recursively re-applying Stage 1 to the sub-regions selected by the scale-space search.

2.3 Stage 2: High-Fidelity Refinement and Generative Modeling

Stage 2 transitions from fast exploration to high-fidelity refinement. It takes the Stage 1 scaffold and a selected characteristic scale τ^* and constructs a generative model of the data manifold at that scale. The central architectural upgrade is to lift the representation from a node–edge graph to a simplicial complex, so that probability mass and geometry are modeled on simplex volumes rather than only on Voronoi cells. This yields a natural objective—*simplex equilibrium*—and a discrete domain on which conservation laws can be enforced for generative modeling.

Stage 2 is run as an intensive refinement pass at fixed τ^* . It warm-starts from Stage 1 (carrying forward node positions, links, and statistics) and initializes a simplicial complex from the existing graph; the initialization and warm-start mechanics are described in SI S9. Learning

then proceeds on a dynamic complex with simplex-native updates and continued topology growth/healing driven by local evidence.

Because a simplicial complex provides only a piecewise-linear (PL) approximation, Stage 2 includes two additional mechanisms that are introduced below as needed. First, a torsion-based geometric audit (the Torsion Ladder) detects localized failures of the PL approximation and proposes targeted structural interventions. Second, a dual-flow reconstruction accumulates face-wise flux evidence and reconstructs a probability field on the complex for likelihood queries and sampling. Where curvature is too strong for a purely PL core, Proteus can optionally attach localized invertible warps; the generative core remains PL in warped coordinates, and the model in the original space is obtained by pullback through the learned warp (SI S7).

2.3.1 The Learning Objective: Simplex Equilibrium

The learning rule in Stage 2 is based on a physical analogy of energy partition. We formalize this by treating the simplex masses as latent variables in a generative model that must explain the observed local transition statistics between mesh nodes (SI S13). **Simplex equilibrium** is then the settled configuration of the mass field at the fixed scale τ^* : the (local) minimum of a free-energy objective that trades off transition prediction error (data fit), prior regularization, and model complexity (SI S3.4). In this view, the error from a data point is an “impulsive force” that increases prediction error locally; the system’s response is determined by its local rigidity: in *plastic* regions (low rigidity), error is reduced by geometric nudges and mass reallocation; in *rigid* regions (high rigidity), error accumulates as structural stress indicating that a discrete edit (split, prune, or a local warp) is needed to lower free energy.

Define Node Equilibrium as the state where the probability mass in each node’s Voronoi cell is approximately equal, $P(V_k) \approx 1/N$ for N nodes. In contrast, Simplex Equilibrium requires equal mass in each simplex, $P(C_i) \approx 1/M$ for M simplices. While not identical, optimizing for Simplex Equilibrium drives the system toward Node Equilibrium via a negative feedback loop. Consider a node i with excess mass $P(V_i) \gg 1/N$. This biases data toward w_i , inflating structural stress in adjacent simplices S_i . The accumulated error vector E_{C_k} grows radially away from w_i , triggering a split that inserts a new node j , redistributing mass from V_i to V_j . Divergences occur at dimensionality junctions (e.g., 1D filament meeting 2D sheet), where bimodal activity and asymmetric links signal complex topology. This “informative mismatch” emerges naturally, providing robust signatures for local structure. A concise correspondence argument, showing that simplex equilibrium induces effective node equilibrium under mild regularity (local stationarity and nondegenerate simplex stars), is provided in SI S12.2. The proof sketches how persistent node over-mass necessarily produces simplex-level stresses whose resolution reduces Voronoi imbalance.

2.3.2 Geometric Audit: Torsion Formalism and the Torsion Ladder

The core innovation of Stage 2 is a rigorous geometric audit designed to address the limitations of the piecewise-linear approximation on a curved manifold. This audit is based on the concept of **torsion**. In our framework, the local residual mean field $\mathbf{m}$ should ideally be conservative (irrotational). Any rotational component in this field indicates a geometric mis-fit that cannot be resolved by simple node translation, signaling a failure in the linear approximation.

We formalize this concept with a discrete 2-form. For a d -simplex $S = \{v_0, \dots, v_d\}$ with vertex positions $\{\mathbf{w}_{v_i}\}$ and residual means $\{\mathbf{m}_{v_i}\}$, we define edge matrices relative to the barycenter: $E = [\mathbf{w}_{v_1} - \bar{\mathbf{w}} \ \cdots \ \mathbf{w}_{v_d} - \bar{\mathbf{w}}]$ and $M = [\mathbf{m}_{v_1} - \bar{\mathbf{m}} \ \cdots \ \mathbf{m}_{v_d} - \bar{\mathbf{m}}]$. The torsion tensor is then given by:

$$\Omega_S = M^\top E - E^\top M.$$

This **simplex 2-form** Ω_S is the direct discrete analogue of the curl ($\nabla \times \mathbf{m}$) of the residual mean field. Its Frobenius norm, $\kappa_S = \|\Omega_S\|_F$, provides a single scalar value for the “torsional

Placeholder schematic: Torsion Ladder triage with thresholds on $R_S = \kappa_S/\tau^*$ and actions (keep, split, mini-NSF).

Figure 1: Torsion Ladder triage with thresholds on $R_S = \kappa_S/\tau^*$ and resulting actions (keep, split, mini-NSF).

stress” within simplex S . A high value of κ_S indicates that the piecewise-linear approximation is failing and that a corrective intervention is necessary.

This audit is operationalized by the **Torsion Ladder**, a triage policy based on the torsion ratio $R_S = \kappa_S/\tau^*$. Intuitively, R_S compares “how curved” the local residual field is to the scale at which the model is currently operating. Low ratios are treated as numerical noise; moderate ratios indicate tolerable curvature and are left unchanged; intermediate ratios trigger additional local settling and, if necessary, a torsion-aligned split that adds discrete capacity and is accepted only when it reduces the local free energy after accounting for complexity (SI S3.4). When curvature is too strong for purely geometric refinement, the ladder escalates to a localized parametric intervention by attaching a small normalizing spline flow (“mini-NSF”) to the affected patch. Default cutoff values for these regimes are provided in the SI (SI S3, SI S7).

To preserve numerical stability, any torsion-aligned split is screened by a simplex shape metric, $Q_S = dr_{\text{in}}/R_{\text{circ}} \in (0, 1]$, where r_{in} and R_{circ} are the inradius and circumradius. If the resulting child simplices would have poor shape quality (default $Q_S < 0.25$), the split is rejected and replaced by a more conservative centroid insertion (a “star split”); details and defaults are in SI S3.

In Stage 2, node pruning is arbitrated by incident simplices: deletions require a formal vote across the simplex star of a node, with a geometric veto to protect structurally critical “tent-pole” nodes (SI S11).

2.3.3 Generative Modeling with the Dual Flow

The framework’s generative capability is enabled by tracking a **dual flow** of probability across simplex faces. For each data point, the residual is projected onto the outward face normals of its containing simplex, producing a signed “pressure” contribution for each face. Aggregating these contributions yields empirical flux evidence on faces; enforcing discrete conservation constraints then reconstructs a conservative face-flux (pressure) field and the associated piecewise-linear probability field on the complex. By the Divergence Theorem, this construction supports both density evaluation and sampling from a single learned representation.

For a winning simplex S and sample $\mathbf{x}$, a fractional face pressure is distributed to each facet $f \in \partial S$ proportional to the projection of the residual onto the outward unit normal $\mathbf{n}_f$:

$$\Delta p_f \propto \max(0, (\mathbf{x} - \bar{\mathbf{w}}_S)^\top \mathbf{n}_f); \quad p_f \leftarrow p_f + \Delta p_f.$$

Let $\mathcal{G}_{\text{dual}}$ be the dual graph whose nodes are faces and whose edges connect adjacent faces. Each primal simplex induces a local flow-consistency constraint on this graph. We seek face pressures $\{p_f\}$ that satisfy discrete conservation (no net source or sink within each simplex, up to sampling noise) while staying close to the accumulated evidence. This yields a quadratic objective over the dual graph:

$$\min_{\{p_f\}} \sum_f \lambda (p_f - \hat{p}_f)^2 + \sum_S \mu \|A_S \mathbf{p}_S\|_2^2,$$

where $\hat{p}_f$ are the empirical tallies, $\mathbf{p}_S$ is the vector of face pressures incident to simplex S , and A_S is a small stencil enforcing local conservation. Solving this quadratic objective on $\mathcal{G}_{\text{dual}}$ yields face pressures $\{p_f\}$ that define a piecewise-linear probability gradient field on the complex; details of the discretization and solver are provided in SI S4.

2.3.4 Structure Learning via Free Energy

Proteus makes discrete topology changes (splits, pruning/merges, and optional warp attachments) using geometric signals as fast proposals and a local evidence test as the arbiter. We define a local region $\mathcal{R}$ affected by a proposed edit as the set of nodes whose simplex stars change under the edit, plus a small neighbor ring to account for router denominators (SI S3.4). Let $\mathcal{L}_{\mathcal{R}}$ denote the transition log-likelihood restricted to sources in $\mathcal{R}$ under the geometry-induced router $q(j \mid i; m)$ (SI S13), and let $\log p(m)$ be a prior on the simplex mass field (e.g., Dirichlet composition or logistic-GMRF smoothness). For each candidate topology M (keep vs. split vs. merge), we warm-start the local mass field from the current solution and settle it to a local MAP configuration $\hat{m}$ while keeping the exterior fixed. The candidate is scored by a free-energy proxy:

$$F(\mathcal{R}; M) = -\mathcal{L}_{\mathcal{R}}(\hat{m}; M) - \log p(\hat{m} \mid M) + \frac{1}{2} k_M \log N_{\text{eff}, \mathcal{R}},$$

where k_M is the number of free parameters introduced in the affected region and $N_{\text{eff}, \mathcal{R}}$ is the effective local transition count. The edit is accepted only if it decreases free energy by a threshold margin, providing a principled justification for non-continuous structure changes under finite data (details in SI S3.4). Crucially, the dual-flow reconstruction above is *conditioned* on the accepted topology and its settled mass/geometry; it does not provide an independent likelihood term for the same mass field, avoiding double-counting of evidence.

2.3.5 Out-of-Sample Queries and Sampling

Once trained, the Proteus representation supports three inference-time uses within a single geometric model: *semantic querying* (e.g., density and anomaly queries, optionally conditioned on constraints), *conditional synthesis* from partial or fuzzy constraints, and *constraint extraction* by aggregating routed mass and transition structure over selected subsets.

Given a query $\mathbf{x}$, locate the containing simplex (or nearest via barycentric projection), compute barycentric coordinates, and evaluate density from face pressures and the piecewise-linear gradient field (see SI S13). For generation, sample a simplex proportional to its mass, draw barycentric coordinates, and map to data space; apply local warps if attached. This enables likelihood queries, anomaly scoring, and synthesis using the same dual-flow field used in training. Boundary faces use Neumann-like treatment (no exterior flux) as in the SI. The procedure is linear in simplex dimension and reuses cached factors.

For evaluation, we report generation metrics (MMD) and log-likelihood as detailed in Section 3 and SI S5.

2.3.6 Non-Linear Warp Strategy (Global vs. Patch-wise)

When residual curvature persists after geometric refinement, Proteus applies a minimal non-linear warp guided by a torsion-coverage statistic P_{κ} (the fraction of simplices whose torsion ratio exceeds a curvature threshold). If curvature is widespread, Proteus trains a shallow global flow (e.g., a lightweight Glow variant) to provide broad non-linear capacity. If curvature is localized, Proteus instead attaches compact mini-NSFs (few-layer rational-quadratic spline couplings) only to the highest-torsion patches, identified via community detection on a facet-adjacency graph. In intermediate cases, a modest global flow can be combined with patch-wise mini-NSFs on the worst regions. If an attached warp fails to reduce torsion in its patch, the ladder falls back to a torsion-aligned split. Default thresholds, patch selection, and training procedures are provided in SI S7.

3 Model Quality Analysis

To validate the claims of the Proteus framework, we conduct a series of experiments designed to measure the fidelity of the generated model, rather than its performance on a specific downstream task.

3.1 Experimental Setup

3.1.1 Datasets

We use a suite of synthetic datasets with known ground-truth topologies, including nested spheres of different dimensions, linked tori, and manifolds with variable density and dimensionality junctions. We also use benchmark real-world datasets to demonstrate performance on complex, unknown structures.

3.1.2 Evaluation Metrics

Our analysis focuses on metrics that directly assess the quality of the learned manifold model:

- **Topological Accuracy:** We compare the Betti numbers (characterizing holes of different dimensions) of the learned simplicial complex to the ground truth of the synthetic datasets using persistent homology.
- **Reconstruction Error:** We measure the average distance between original data points and their projection onto the learned manifold.
- **Log-Likelihood:** As Proteus produces a generative model, we evaluate the average log-likelihood of held-out test data under the model. This provides a direct measure of its predictive power, a capability absent in non-generative manifold learning methods.
- **Generative Quality:** We draw samples from the learned model and compare their distribution to the original data’s distribution using metrics such as Maximum Mean Discrepancy (MMD) to assess the fidelity of the generative process.

We follow standardized visualization and protocol defaults detailed in SI S15 (visuals) and SI S5 (evaluation settings).

3.2 Comparative Analysis

We will present results (tables, visualizations) comparing the Proteus model’s performance on the above metrics to models produced by other state-of-the-art algorithms, such as UMAP (for embedding quality) and a Variational Autoencoder (for generative quality).

3.2.1 Protocol Details

- **Datasets:** Synthetic: nested spheres, linked tori, variable-density manifolds; Real-world: standard benchmarks (e.g., MNIST subset, tabular with mixed scales).
- **Splits & seeds:** 80/20 train/test; 5 seeds per experiment; report mean $\pm$ std.
- **Scales grid:** Geometric grid in τ : $\tau_i = \tau_0 r^i$, $r=0.71$; 7–9 points per decade; GP refinement within first bracketed maximum.
- **Stopping:** Stage 1: $CV < 0.01$; Stage 2: $CV < 0.01$ and $R_S < 0.30$ for all simplices.
- **Persistent Homology:** Vietoris–Rips with max dim ≤ 2 ; filtration up to $1.5\sigma_*$; bottle-neck/Wasserstein distances to truth.

Table 1: Placeholder results: topological accuracy (PH distance), reconstruction error, log-likelihood, and MMD (lower is better where applicable).

Method	PH Dist.	Recon Err	Log-Lik	MMD
Proteus (full)	—	—	—	—
No Stage 1	—	—	—	—
No Ladder	—	—	—	—
No Dual Flow	—	—	—	—
Baseline (UMAP+KDE)	—	—	—	—
VAE	—	—	—	—

Placeholder ablation plots: effects of removing Stage 1 scaffold, Torsion Ladder, or Dual Flow on PH distance, reconstruction error, and MMD across datasets.

Figure 2: Ablations (placeholders): effects of removing Stage 1 scaffold, Torsion Ladder, or Dual Flow on PH distance, reconstruction error, and MMD across datasets.

- **MMD:** RBF kernel with bandwidth median heuristic; 1 000 samples; 1 000 test points.
- **Ablations:** (i) No Stage 1 scaffold; (ii) No Torsion Ladder; (iii) No Dual Flow.

3.3 Ablation Studies

We conduct targeted ablation studies to demonstrate the necessity of the core components of the Proteus architecture:

- **Ablating Stage 1 Scaffolding:** We run the Stage 2 refinement process from a random initialization instead of the Stage 1 scaffold. We expect to show that this leads to significantly slower convergence, higher final reconstruction error, and frequent failure to identify the correct global topology, demonstrating the critical importance of the coarse-to-fine, scale-aware approach.
- **Ablating the Torsion Audit:** We disable the Torsion Ladder, forcing the model to rely only on variance-based splitting. On datasets with non-trivial curvature, we expect the model to produce topologically incorrect structures and exhibit high reconstruction error in curved regions, validating the need for a direct geometric audit.
- **Ablating the Dual Flow:** We disable the dual flow mechanism and evaluate the model’s generative capabilities. We expect to show that without the explicit modeling of the probability gradient, the model’s ability to produce high-fidelity samples and provide accurate density estimates is severely degraded.

4 Limitations and Complexity/Scaling

Limitations: (i) Extremely high ambient dimensions without an initial projection can inflate ANN and per-sample costs; (ii) Highly anisotropic densities may require a deeper grid in τ to separate overlapping scales; (iii) In very sparse data, torsion estimates on large simplices can be noisy—guarded by our Q_S fallback and minimum-sample thresholds but still a practical consideration.

Typical envelopes (indicative):

- $N \in [10^5, 10^7]$, $d \in [64, 2048]$ after optional projection; per-level wall-time dominated by HNSW queries and moment updates.

- HNSW: $M \in [16, 32]$, $\text{efConstruction} \in [100, 400]$, $\text{efSearch} \in [50, 200]$; increase efSearch near convergence for stability.
- Memory: nodes+links+simplexes typically within $\sim 10\%$ growth per recursion level due to Ladder and prune rules.

Asymptotic envelopes: Stage 1 per-epoch time is $\mathcal{O}(N d k \log N_{\text{nodes}})$ with HNSW queries; topology/pruning is amortized. Space is $\mathcal{O}(N_{\text{nodes}} d + N_{\text{nodes}} \bar{d})$, independent of N . Stage 2 seeding via Greedy Chaining costs $\mathcal{O}(N_c d_{\text{cand}}^2 d_c)$; periodic torsion audits cost $\mathcal{O}(N_{\text{simplices}} (d+1)^2 d)$. Dual-flow belief propagation operates on the dual graph of faces of size $\mathcal{O}(N_{\text{simplices}} d)$ and converges in near-linear time per pass.

Complexity/Scaling: Per-sample compute is $\mathcal{O}(k d)$; deferred moves/splits trigger occasional $\mathcal{O}(d^2)$ work. Stage 1’s grid uses 7–9 points per decade with GP refinements (few evaluations), leading to wall-time dominated by ANN queries and moment updates; HNSW scales well sub-linearly with N . Stage 2 adds simplex-native operations; torsion and shape metrics are $\mathcal{O}(d^2)$ but applied sparsely on topology events. Overall, memory grows with nodes/links/simplex count; empirical growth stays within a $\sim 10\%$ budget per recursion level due to the Torsion Ladder and prune rules.

5 Conclusion

Proteus targets a common practical reality in modern manifold learning: progress is often achieved by stitching together a toolchain of embeddings, topological diagnostics, density estimators, and geometric heuristics, each providing only part of the desired modeling stack. Building on competitive Hebbian learning’s maximum-entropy partitioning, Proteus integrates fast multiscale scaffold discovery with evidence-grounded refinement into a simplicial generative core. The two-stage, scale-aware design allows Proteus to discover candidate structure efficiently and then allocate refinement capacity only where supported by finite-data evidence. Our analysis shows that this hybrid approach produces models that are not only topologically faithful but also support likelihood-based inference and high-fidelity generation within a single representation.

A key advantage of Proteus is that many quantities are *derived from first principles* rather than tuned by hand: the geometric grid ratio r , the EWMA weight α , and the dual motion rates η_{cent} and η_{GNG} all follow from scale-space resolution, statistical half-life, and variance-correction arguments (see SI S2). Together with the statistical gauntlets for pruning (SI S3), these choices make the system robust, interpretable, and largely parameter-free in practice.

This work establishes a robust method for modeling complex data. The rich, generative model produced by the Proteus pipeline serves as a foundational layer for a wide array of subsequent applications, from high-performance search to automated pattern discovery. These applications, along with architectural enhancements for streaming data, will be detailed in subsequent work.

As shown in ablations, the two-stage design and torsion audit are necessary for both topological fidelity and generative quality; the added discussion on Simplex vs. Node Equilibria explains why our simplex-level objective still drives node-level balance in practice.

6 References Prep

This section collects potential references to be formalized and cited.

Manifold Learning: Isomap (Tenenbaum et al., 2000); Locally Linear Embedding (Roweis & Saul, 2000); t-SNE (van der Maaten & Hinton, 2008); UMAP (McInnes, Healy, Melville, 2018).

Hierarchical and Density Clustering: HDBSCAN (Campello, Moulavi, Sander, 2013/2015).

Scale-Space Theory: Lindeberg, Scale-Space Theory (1994, 1998).

Growing Neural Gas and Variants: GNG (Fritzke, 1995) and node-centric extensions.

Discrete Geometry and Topology: Discrete exterior calculus / discrete differential forms (Desbrun et al.); Divergence Theorem references (standard vector calculus texts); Persistent homology and Betti numbers (Edelsbrunner & Harer).

Generative Models: VAEs (Kingma & Welling, 2013/2014); GANs (Goodfellow et al., 2014); Normalizing Flows (Dinh et al., 2014/2016; Papamakarios et al., 2021).

Appendix: Notation

We summarize the core symbols used throughout the paper.

- $m_i \in \mathbb{R}^d$: EWMA residual mean at node i .
- $s_i \in \mathbb{R}^d$: EWMA of squared residual at node i ; variance $\sigma_i^2 = s_i - m_i^2$.
- $\rho_i = \|m_i\|/(\sigma_i + \varepsilon)$: incoherence ratio at node i .
- k : neighborhood size (Stage 1: $k=8$; Stage 2: $k \approx d$).
- $D_{subspace}$: current working dimensionality; $d_{final,i}$: smoothed local intrinsic dimension at node i .
- $s_{control} \in [0, 1)$: optimizer control; $\tau_{global} = -D_{subspace} \log(1-s_{control})$.
- $\tau_{local,i} = -d_{final,i} \log(1-s_{control})$: per-node growth threshold.
- $\Phi(\tau)$: scale-space response evaluated at threshold τ .
- Ω_S : simplex 2-form (discrete torsion) on simplex S ; $\kappa_S = \|\Omega_S\|_F$.
- $R_S = \kappa_S/\tau^*$: torsion ratio used by the Torsion Ladder at the fixed scale τ^* .
- $\tilde{\rho}$: neighbor-normalized incoherence; $CV = \text{std}(\tilde{\rho})/\text{mean}(\tilde{\rho})$.